No.of Functional Features Included in the Solution

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the OptiCrop: Smart Agricultural Production Optimization Engine. The application utilizes machine learning algorithms and environmental data to recommend suitable crops based on soil nutrients and climatic conditions. Each feature contributes to improving agricultural productivity, resource utilization, and decision-making for farmers.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Module	Accepts Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall values	User Interface	Done	High
2	Data Preprocessing	Cleans and validates user input before prediction	Data Processing	Done	High
3	Crop Recommendation	Predicts the most suitable crop using Machine Learning	ML Prediction Model	Done	High
4	Flask Web Interface	Provides an interactive web application for users	Frontend & Backend	Done	Medium
5	Model Integration	Integrates the trained ML model with the Flask application	Backend	Done	High
6	Prediction Result Display	Displays the recommended crop with user-friendly output	User Interface	Done	Medium
7	Input Validation	Prevents invalid or incomplete user inputs	Validation Module	Done	Medium
8	Error Handling	Handles application errors and invalid requests gracefully	Backend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Prediction Result Display
2	Backend / Logic	3	Input Validation, Business Logic, Flask Routing
3	Database / Storage	1	Dataset Management using CSV/Pandas
4	API / Integration	1	Machine Learning Model Integration
5	Security / Authentication	1	Input Validation and Error Handling